

FINAL PROJECT REPORT

SUBMITTED BY

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COLLEGE NAME

**DR. D. Y. PATIL PRATISHTHAN'S COLLEGE OF ENGINEERING, SALOKHENAGAR,
KOLHAPUR**

BRANCH

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

ROLL ID

B.TECH 46

PROJECT NAME

India's Agricultural Crop Production Analysis (1997–2021)

Final Project Report Template

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Final Project Report

1. Introduction

1.1 Project Overview

The **India's Agricultural Crop Production Analysis (1997–2021)** project analyzes agricultural crop production in India using Tableau. The project converts raw agricultural data into interactive visualizations and dashboards.

The analysis focuses on **state-wise, district-wise, crop-wise, season-wise, area-wise, production-wise, and year-wise** agricultural patterns and trends from 1997 to 2021.

1.2 Objectives

- To analyze agricultural crop production in India.
- To compare crop production across states and districts.
- To identify major crops based on production.
- To analyze production across different seasons.
- To study year-wise crop production trends.
- To analyze the relationship between cultivated area and production.
- To develop an interactive Tableau dashboard.
- To present meaningful agricultural insights through a Tableau Story.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An agricultural policymaker	Understand crop production across different states and districts.	It is difficult to compare production levels between different regions.	Agricultural production varies across states, districts, crops, seasons, and years.	Uncertain about where agricultural support and resources are most needed.
PS-2	A farmer	Understand crop production patterns across different crops and years.	It is difficult to compare crop production using raw data.	Production varies based on crop type, season, location, and cultivated area.	Unsure about important production trends.
PS-3	An agricultural analyst	Analyze crop production trends.	The dataset contains a large number of records across states, districts, crops, seasons, and years.	Large datasets make manual analysis difficult.	It is difficult to quickly identify important agricultural patterns.

2.2 Project Proposal – Proposed Solution

Section	Details
Objective	Analyze agricultural crop production in India from 1997 to 2021 using Tableau and identify important production patterns and trends.
Scope	State-wise, district-wise, crop-wise, season-wise, area-wise, production-wise, and year-wise analysis.
Problem Statement	Agricultural stakeholders need an easy way to understand and compare crop production across different regions, crops, seasons, and years.
Impact	The analysis can help identify production patterns, regional differences, major crops, and historical trends for better agricultural planning.
Section	Details
Approach	Collect, clean, prepare, analyze, and visualize the agricultural dataset using Tableau.
Key Features	KPI cards, charts, maps, filters, crop comparison, state-wise analysis, district-wise analysis, trend analysis, interactive dashboard, and Tableau Story.

2.3 Initial Project Planning

Sprint	Epic	User Story	Story Points	Priority	Team	Start Date	End Date
Sprint-1	Data Collection & Cleaning	USN-1- Collect, understand, and clean the agricultural dataset.	3	High	Self	17Aug2026	18Aug2026
Sprint-1	Tableau Data Preparation	USN-2 – Prepare and connect the dataset to Tableau.	2	High	Self	18Aug2026	19Aug2026
Sprint-2	AI Adoption Analysis	USN-3 - Analyzecrop production across states and districts.	3	High	Self	19Aug2026	20Aug2026
Sprint-2	Student & Teacher Analysis	USN-4 – Compare crop and seasonal production.	3	High	Self	19Aug2026	20Aug2026
Sprint-2	Dashboard Development	USN-5 – Develop an interactive Tableau dashboard.	5	High	Self	20Aug2026	21Aug2026
Sprint-2	Testing & Reporting	USN-6 – Test the dashboard and prepare insights.	3	High	Self	21Aug2026	22Aug2026

3. Data Collection and Pre-processing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Section	Details
---------	---------

Project Overview	The project analyzes agricultural crop production in India from 1997 to 2021 using the project-provided agricultural dataset.
Data Collection Plan	The agricultural crop production dataset was collected in Excel format and prepared for analysis and visualization in Tableau..
Raw Data Source	India’s Agricultural Crop Production Dataset
Format	Excel (.xlsx)
Access	Project-provided / Public dataset

Raw Data Source Screenshot

Raw Dataset

1	State	District	Crop	Year	season	Area	Area Units	Production	Production Units	Production Units
2	Andaman and Nicobar Islands	NICOBARS	Arecanut	2001-02	Kharif	1254	Hectare	2061	Tonnes	1.64354067
3	Andaman and Nicobar Islands	NICOBARS	Arecanut	2002-03	Whole Year	1258	Hectare	2083	Tonnes	1.655802862
4	Andaman and Nicobar Islands	NICOBARS	Arecanut	2003-04	Whole Year	1261	Hectare	1525	Tonnes	1.209357653
5	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAM	Arecanut	2001-02	Kharif	3100	Hectare	5239	Tonnes	1.69
6	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut	2002-03	Whole Year	3105	Hectare	5267	Tonnes	1.696296296
7	Andaman and Nicobar Islands	SOUTH ANDAMANS	Arecanut	2003-04	Whole Year	3118	Hectare	5182	Tonnes	1.661962797
8	Andaman and Nicobar Islands	NICOBARS	Banana	2002-03	Whole Year	213	Hectare	1278	Tonnes	6
9	Andaman and Nicobar Islands	NICOBARS	Banana	2003-04	Whole Year	266	Hectare	1763	Tonnes	6.627819549
10	Andaman and Nicobar Islands	SOUTH ANDAMANS	Banana	2002-03	Whole Year	1524	Hectare	10882	Tonnes	7.140419948
11	Andaman and Nicobar Islands	SOUTH ANDAMANS	Banana	2003-04	Whole Year	1530	Hectare	11558	Tonnes	7.554248366
12	Andaman and Nicobar Islands	NICOBARS	Black pepper	2002-03	Whole Year	63	Hectare	13.5	Tonnes	0.214285714
13	Andaman and Nicobar Islands	NICOBARS	Black pepper	2003-04	Whole Year	75.5	Hectare	15.86	Tonnes	0.210066225
14	Andaman and Nicobar Islands	SOUTH ANDAMANS	Black pepper	2002-03	Whole Year	487	Hectare	102.5	Tonnes	0.210472279
15	Andaman and Nicobar Islands	SOUTH ANDAMANS	Black pepper	2003-04	Whole Year	497	Hectare	104.37	Tonnes	0.21
16	Andaman and Nicobar Islands	NICOBARS	Cashewnut	2001-02	Whole Year	719	Hectare	192	Tonnes	0.267037552
17	Andaman and Nicobar Islands	NICOBARS	Cashewnut	2002-03	Whole Year	719	Hectare	208	Tonnes	0.289290682
18	Andaman and Nicobar Islands	NICOBARS	Cashewnut	2003-04	Whole Year	717	Hectare	208.5	Tonnes	0.290794979
19	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAM	Cashewnut	2001-02	Whole Year	81	Hectare	33	Tonnes	0.407407407
20	Andaman and Nicobar Islands	SOUTH ANDAMANS	Cashewnut	2002-03	Whole Year	81	Hectare	24	Tonnes	0.296296296
21	Andaman and Nicobar Islands	SOUTH ANDAMANS	Cashewnut	2003-04	Whole Year	116.5	Hectare	26.14	Tonnes	0.224377682
22	Andaman and Nicobar Islands	NICOBARS	Coconut	2001-02	Whole Year	18190	Hectare	64430000	Nuts	3542.056075
23	Andaman and Nicobar Islands	NICOBARS	Coconut	2002-03	Whole Year	18240	Hectare	67490000	Nuts	3700.109649
24	Andaman and Nicobar Islands	NICOBARS	Coconut	2003-04	Whole Year	18284.74	Hectare	68580000	Nuts	3750.66859
25	Andaman and Nicobar Islands	NORTH AND MIDDLE ANDAM	Coconut	2001-02	Whole Year	7015	Hectare	25250000	Nuts	3599.429793
26	Andaman and Nicobar Islands	SOUTH ANDAMANS	Coconut	2002-03	Whole Year	7060	Hectare	26830000	Nuts	3800.283286
27	Andaman and Nicobar Islands	SOUTH ANDAMANS	Coconut	2003-04	Whole Year	7110	Hectare	26660000	Nuts	3749.648383
28	Andaman and Nicobar Islands	NICOBARS	Dry chillies	2002-03	Whole Year	413	Hectare	28.8	Tonnes	0.069733656
29	Andaman and Nicobar Islands	NICOBARS	Dry chillies	2003-04	Whole Year	60	Hectare	102	Tonnes	1.7
30	Andaman and Nicobar Islands	SOUTH ANDAMANS	Dry chillies	2002-03	Whole Year	239	Hectare	20.9	Tonnes	0.087447699
31	Andaman and Nicobar Islands	SOUTH ANDAMANS	Dry chillies	2003-04	Whole Year	340	Hectare	578	Tonnes	1.7

3.2 Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
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Agricultural Crop Production Dataset	Missing or blank values	Moderate	Identify and appropriately handle missing values.
Agricultural Crop Production Dataset	Duplicate records	Moderate	Check and remove duplicate records.
Agricultural Crop Production Dataset	Inconsistent category names	Moderate	Standardize State, District, Crop, and Season values.
Agricultural Crop Production Dataset	Incorrect data types	Low	Convert fields to appropriate Tableau data types.
Agricultural Crop Production Dataset	Unnecessary fields	Low	Remove or exclude fields not required for analysis.
Agricultural Crop Production Dataset	Unusual values	Moderate	Check and validate unusual Area and Production values.

Raw Data Source

Attribute	Description
Dataset Name	India's Agricultural Crop Production Dataset
File Format	Excel (.xlsx)
Number of Records	345,407

Number of Attributes	10
Time Period	1997–2021
Analysis Tool	Tableau

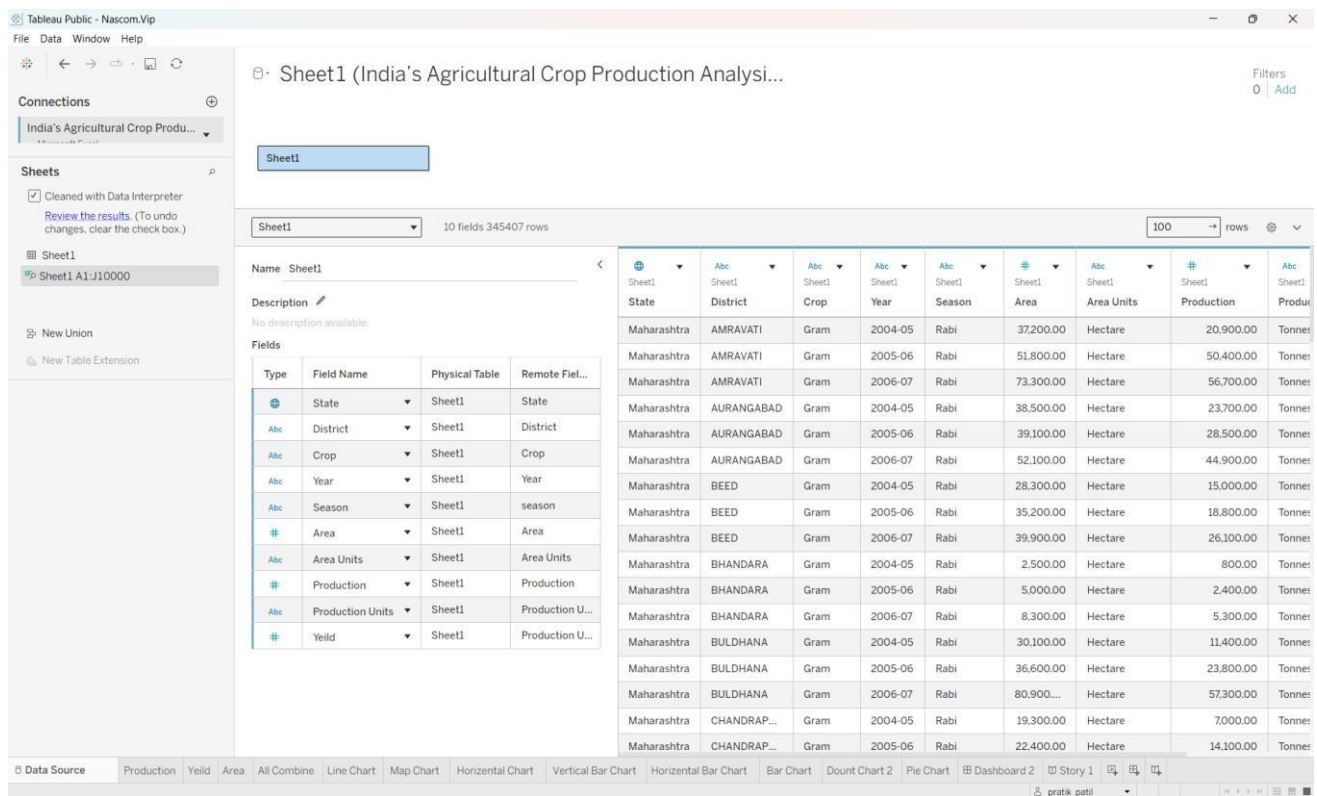
Dataset Attributes

The dataset contains agricultural information including:

- State
- District
- Crop
- Year
- Season
- Area
- Production
- Unit-related information
- Crop production-related fields

Data Quality Screenshot

Data Cleaning / Data Quality



3.3 Data Exploration and Pre-processing

Section	Description
---------	-------------

Data Overview	Dataset contains agricultural crop production information across states, districts, crops, seasons, years, area, and production.
Data Cleaning	Missing values, duplicates, inconsistent values, and unnecessary fields were checked.
Data Transformation	Data was filtered, sorted, and prepared for Tableau analysis.
Data Type Conversion	. Numerical, text, and year fields were assigned appropriate data types.
Column Splitting/Merging	Fields were reviewed and organized where required.
Data Modeling	Dimensions and measures were prepared for Tableau visualizations and calculations.
Save Processed Data	The processed data was connected to and saved within the Tableau workbook.

Pre-processing Screenshot

Tableau Public - Nascom.Vip

File Data Window Help

Connections

India's Agricultural Crop Produ...

Enter sheet name

Sheets

☒ Cleaned with Data Interpreter

Review the results. (To undo changes, clear the check box.)

Sheet1

Sheet1 A1:J10000

New Union

New Table Extension

Sheet1 (India's Agricultural Crop Production Analysis...)

Filters: 0 | Add

Sheet1

Logical Table: Sheet1

Sheet1 10 fields 345407 rows 100 rows

Name: Sheet1

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
State	State	Sheet1	State
Distric	District	Sheet1	District
Crop	Crop	Sheet1	Crop
Year	Year	Sheet1	Year
Season	Season	Sheet1	Season
Area	Area	Sheet1	Area
Area Units	Area Units	Sheet1	Area Units
Production	Production	Sheet1	Production
Production Units	Production Units	Sheet1	Production U...
Yield	Yield	Sheet1	Production U...

State	District	Crop	Year	Season	Area	Area Units	Production	Production Units
Maharashtra	AMRAVATI	Gram	2004-05	Rabi	37,200.00	Hectare	20,900.00	Tonnes
Maharashtra	AMRAVATI	Gram	2005-06	Rabi	51,800.00	Hectare	50,400.00	Tonnes
Maharashtra	AMRAVATI	Gram	2006-07	Rabi	73,300.00	Hectare	56,700.00	Tonnes
Maharashtra	AURANGABAD	Gram	2004-05	Rabi	38,500.00	Hectare	23,700.00	Tonnes
Maharashtra	AURANGABAD	Gram	2005-06	Rabi	39,100.00	Hectare	28,500.00	Tonnes
Maharashtra	AURANGABAD	Gram	2006-07	Rabi	52,100.00	Hectare	44,900.00	Tonnes
Maharashtra	BEEED	Gram	2004-05	Rabi	28,300.00	Hectare	15,000.00	Tonnes
Maharashtra	BEEED	Gram	2005-06	Rabi	35,200.00	Hectare	18,800.00	Tonnes
Maharashtra	BEEED	Gram	2006-07	Rabi	39,900.00	Hectare	26,100.00	Tonnes
Maharashtra	BHANDARA	Gram	2004-05	Rabi	2,500.00	Hectare	800.00	Tonnes
Maharashtra	BHANDARA	Gram	2005-06	Rabi	5,000.00	Hectare	2,400.00	Tonnes
Maharashtra	BHANDARA	Gram	2006-07	Rabi	8,300.00	Hectare	5,300.00	Tonnes
Maharashtra	BULDHANA	Gram	2004-05	Rabi	30,100.00	Hectare	11,400.00	Tonnes
Maharashtra	BULDHANA	Gram	2005-06	Rabi	36,600.00	Hectare	23,800.00	Tonnes
Maharashtra	BULDHANA	Gram	2006-07	Rabi	80,900.00	Hectare	57,300.00	Tonnes
Maharashtra	CHANDRAPUR	Gram	2004-05	Rabi	19,300.00	Hectare	7,000.00	Tonnes
Maharashtra	CHANDRAPUR	Gram	2005-06	Rabi	22,400.00	Hectare	14,100.00	Tonnes

Data Source: Production Yield Area All Combine Line Chart Map Chart Horizontal Chart Vertical Bar Chart Horizontal Bar Chart Bar Chart Dount Chart 2 Pie Chart Dashboard 2 Story 1 pratik patil

4. Data Visualization

4.1 Framing Business Questions

The following business questions were developed from the Tableau workbook:

1. What is the overall agricultural production in India?
2. Which states have higher agricultural production?

3. What is the trend of agricultural yield over the years?
4. How does cultivated area vary across different crops?
5. How does cultivated area compare across states?
6. Which state has the highest agricultural production?
7. How does total agricultural production change year-wise?
8. How is production distributed by unit type?
9. What is the production share across different seasons?

4.2 Developing Visualizations

Visualization 1

Business Question: What is the overall agricultural production in India?

Visualization: India Agricultural Summary **Attributes:**
Area, Production, Yield

Screenshot:

India Agriculture summary

Yeild	27.43M
Area	4,031.13M
Production	330,997.84M

Visualization 2

Business Question:

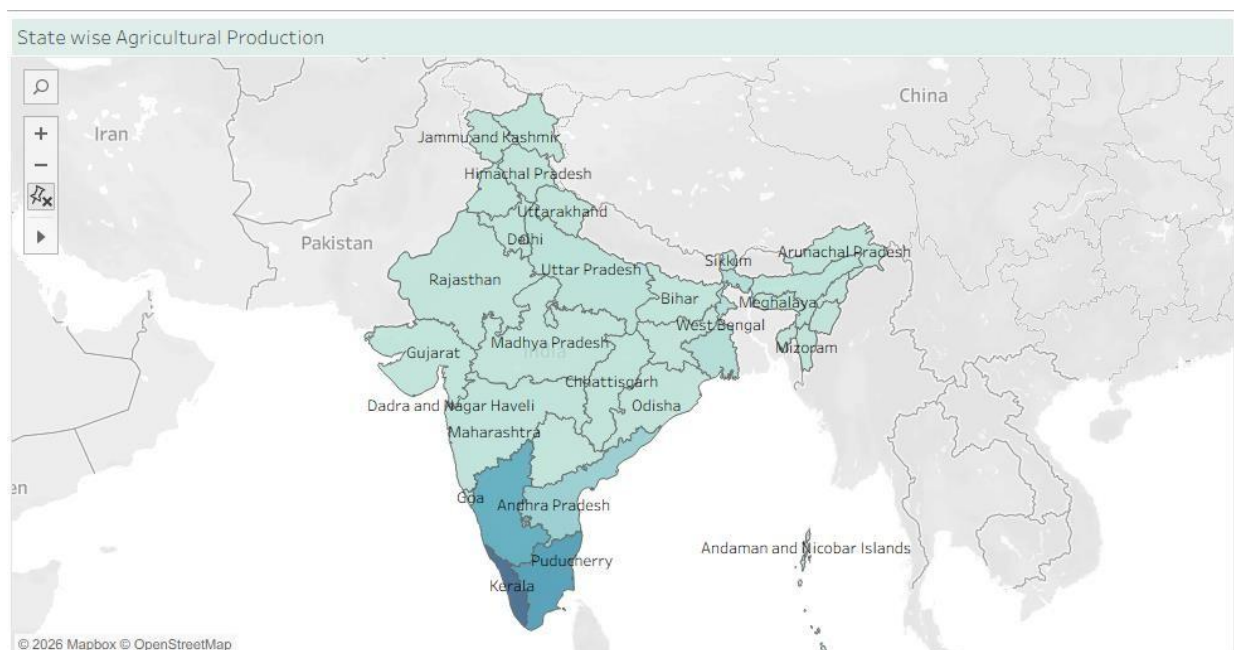
Visualization 2

Business Question: Which states have higher agricultural production?

Visualization: State-wise Agricultural Production – Map

Attributes: State, Production

Screenshot:



Business Question: What is the trend of agricultural yield over the years?

Visualization: Trend of Agricultural Yield – Line Chart

Attributes: Year, Yield

Visualization 3

Screenshot:



Visualization 4

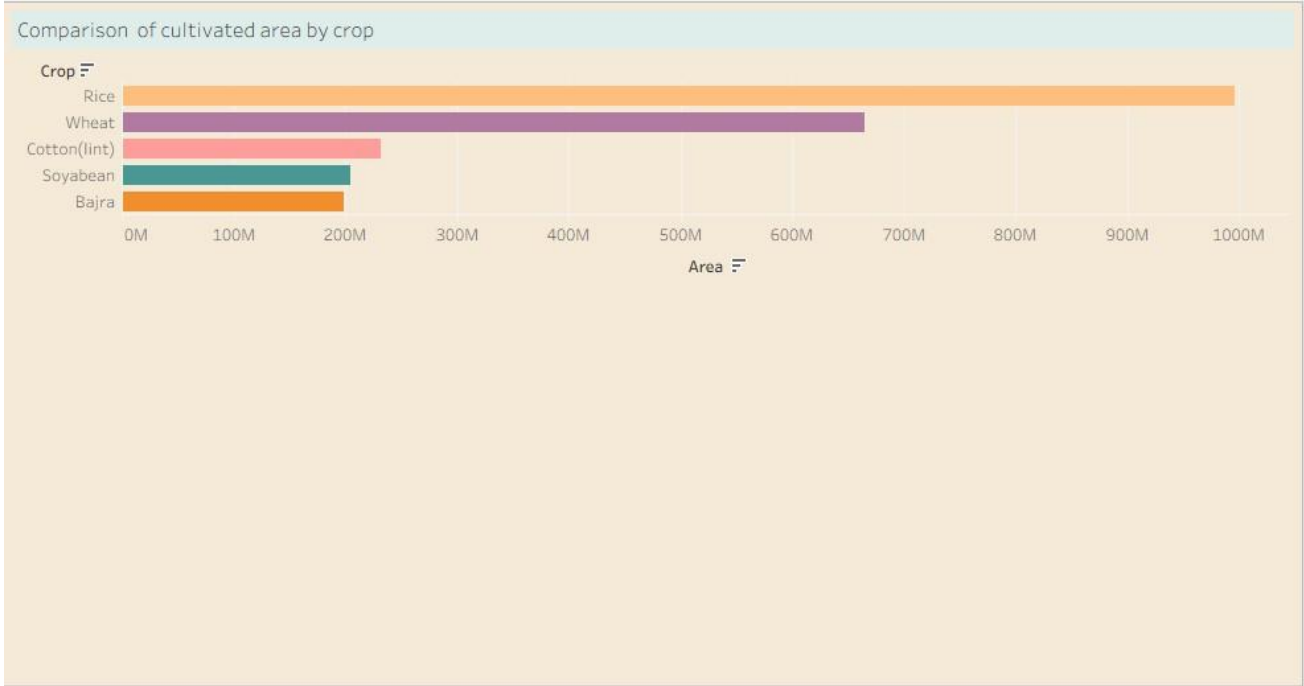
Business Question:

Business Question: How does cultivated area vary across different crops?

Visualization: Comparison of Cultivated Area by Crop _ Horizontal Bar Chart

Attributes: Crop, Area

Screenshot:



Visualization 5

Business Question: How does cultivated area compare across states?
Visualization: State-wise Cultivated Area Comparison – Vertical Bar Chart
Attributes: State, Area



Visualization 6

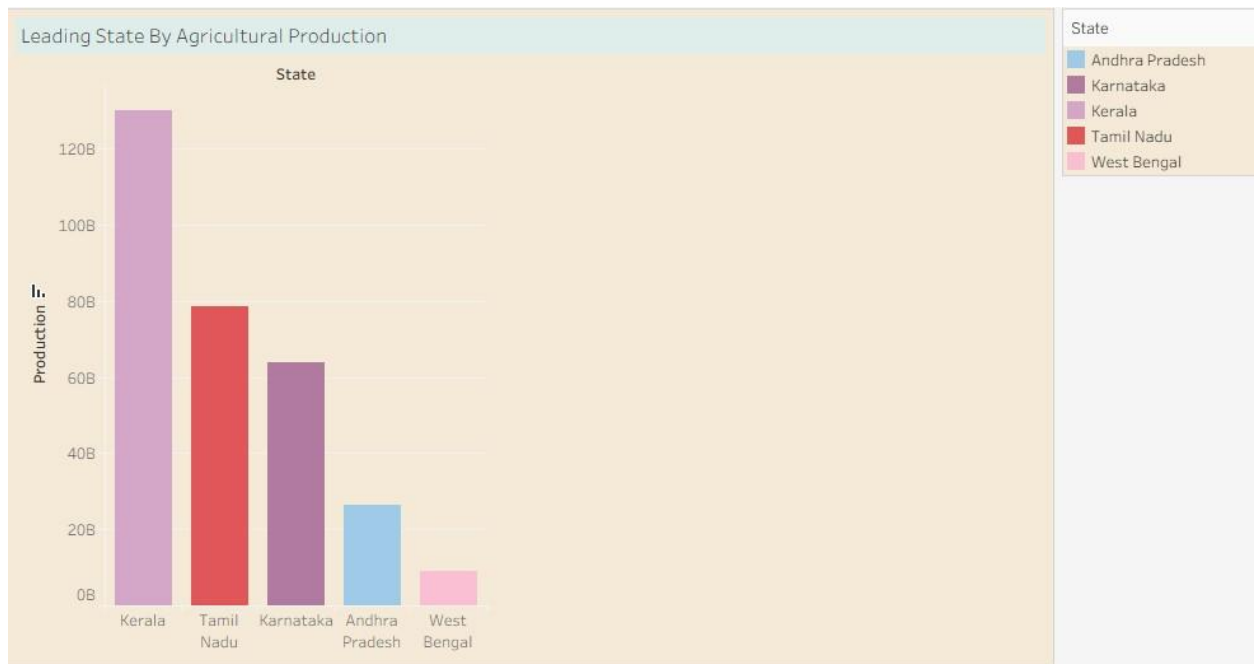
Business Question:

Business Question: Which state has the highest agricultural production?

Visualization: Leading State by Agricultural Production – Horizontal Bar Chart

Attributes: State, Production

Screenshot:



Visualization 7

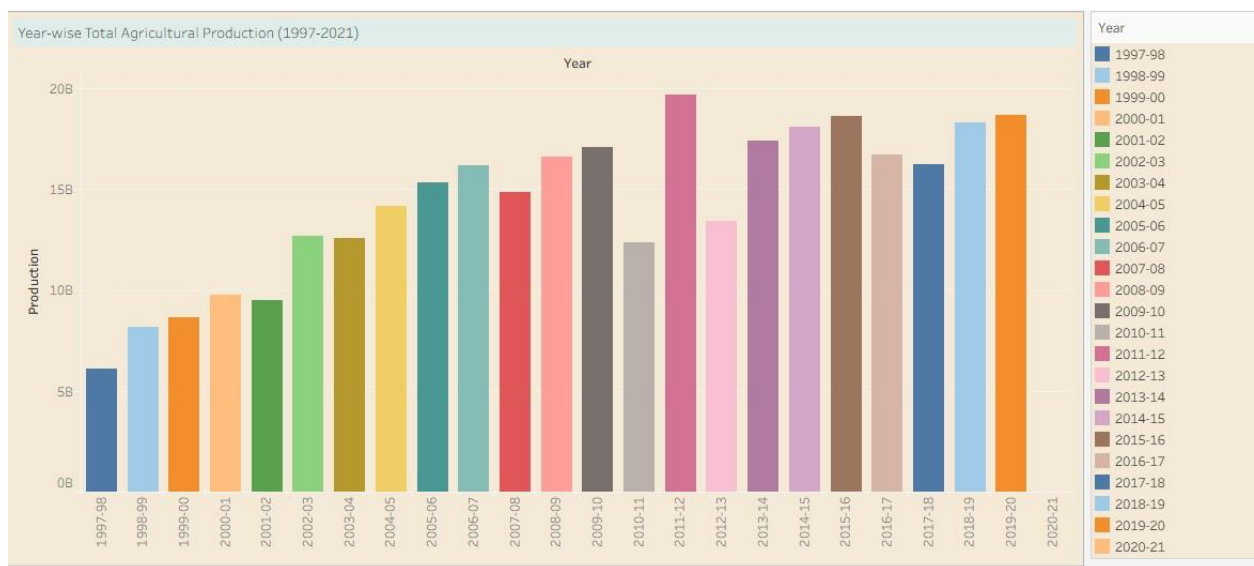
Business Question:

Business Question: How does total agricultural

production change year-wise?

Visualization: Year-wise Total Agricultural

Attributes: Year, Production



Visualization 8

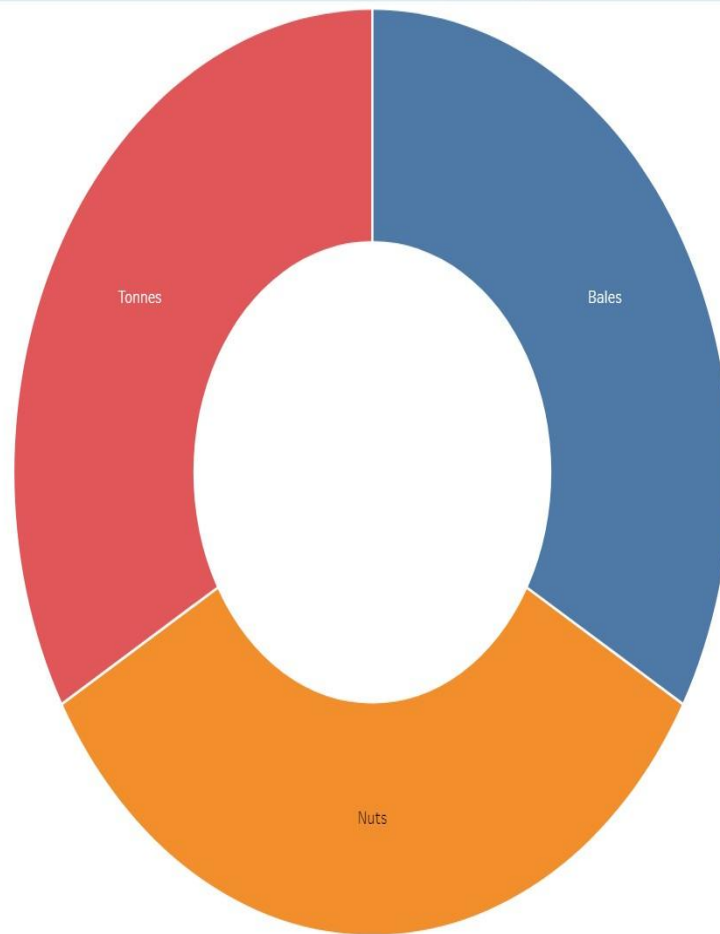
Business Question:

Business Question: How is production distributed by unit type?

Visualization: Production Distribution by Unit Type – Donut Chart

Attributes: Unit Type, Production

Production Distribution by Unit Type



Screenshot:

Visualization 9

Business Question:

Business Question: What is the production

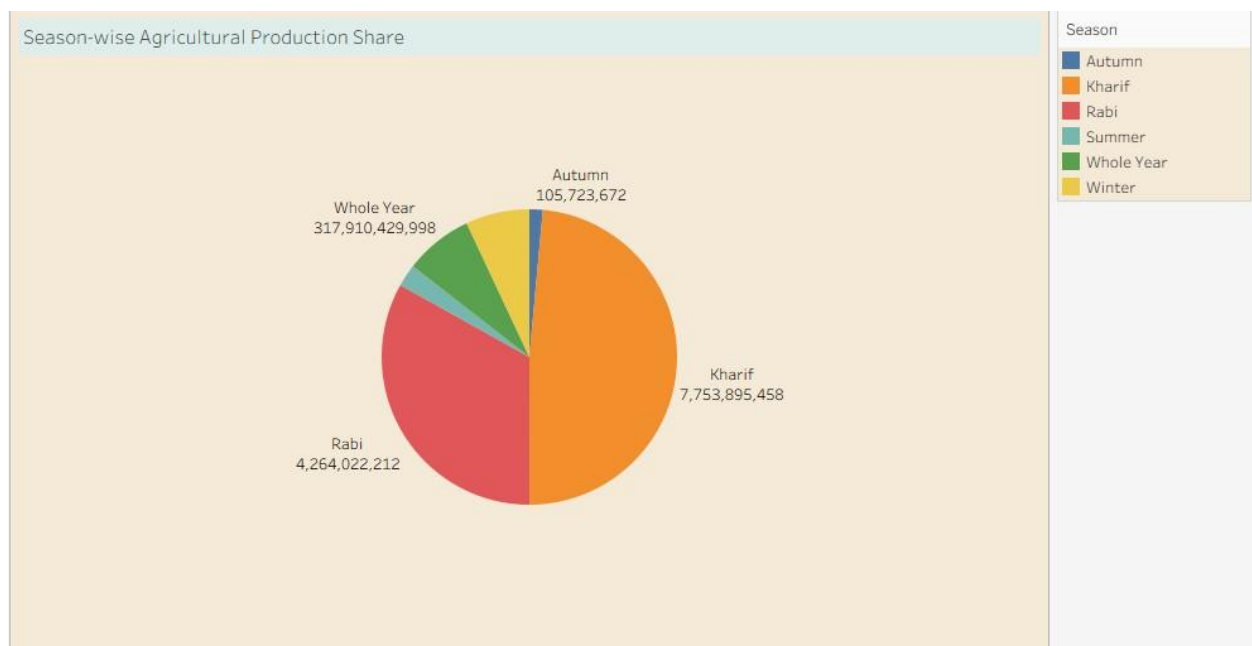
share across different seasons?

Visualization:Season-wise Agricultural

Production Share – Pie Chart

Attributes: Season, Production

Screenshot:



5. Dashboard

5.1 DASHBOARD DESIGN FILE

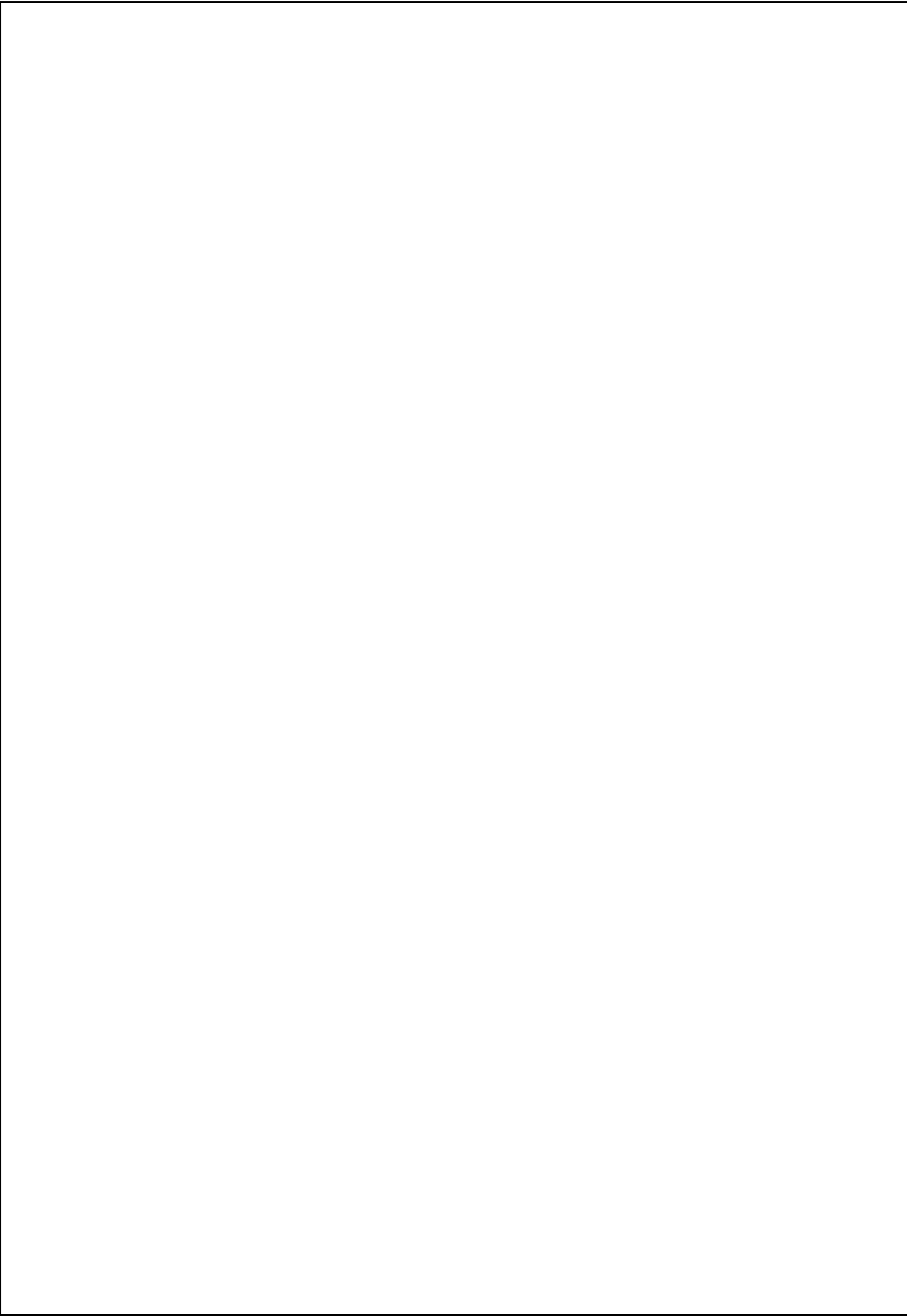
The dashboard combines **KPI cards, charts, maps, filters, and comparison visualizations** to provide an interactive overview of India's agricultural crop production from 1997 to 2021.

Dashboard Design Principles

- Clear and intuitive layout
- Appropriate visualization selection
- Consistent colour and theme
- Interactive filters
- Geographical visualization
- KPI cards
- Comparison charts
- Trend analysis
- Interactive dashboard navigation

Key Features

1. Interactive Tableau Dashboard
2. KPI Cards
3. State-wise Analysis
4. District-wise Analysis
5. Crop-wise Comparison
6. Season-wise Analysis



7. Year-wise Trends
8. Area & Production Analysis
9. Interactive Filters
10. Comparison Charts
11. Key Agricultural Insights

MAIN TABLEAU DASHBOARD

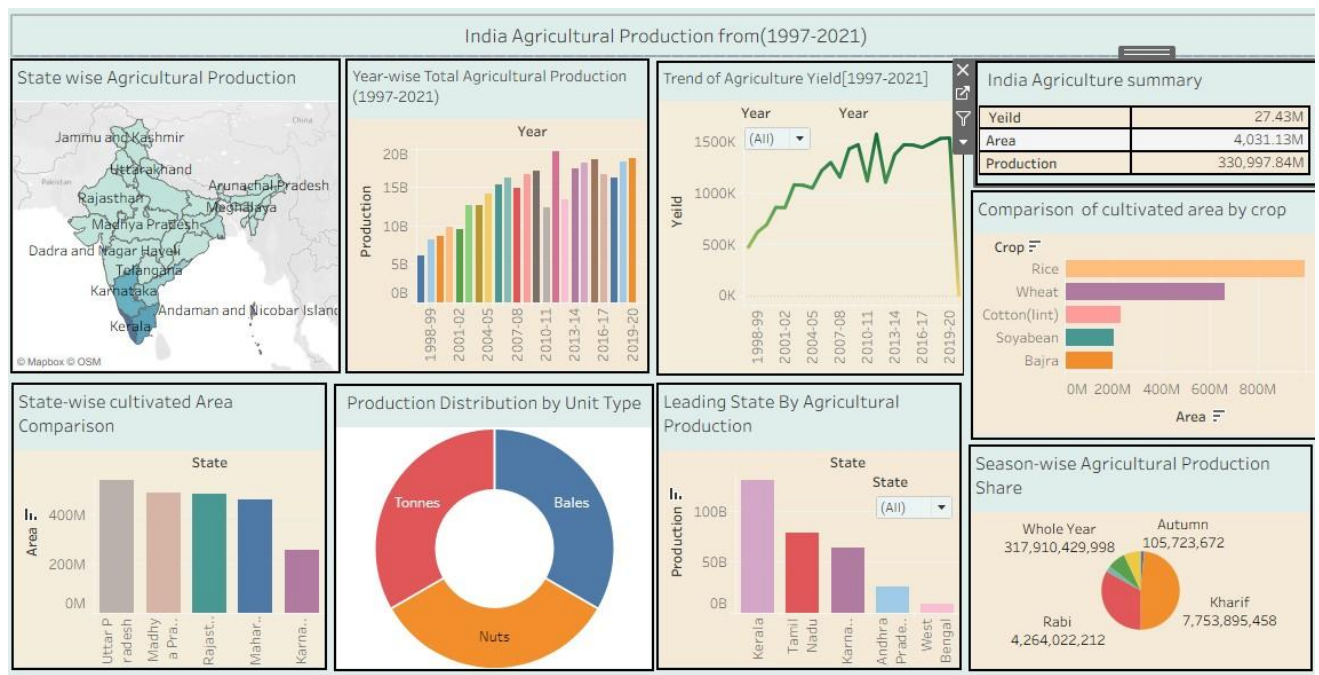


Tableau Public Dashboard Link

Link:https://public.tableau.com/views/Book2_17877409344640/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report

6.1 Story Design File

The Tableau Story presents the findings in a logical sequence, beginning with overall agricultural production and progressing to state, district, crop, season, area, and year-wise analysis.

STORY FLOW

Story Point 1 → India Agricultural Summary Overall Area, Production, and Yield using KPI cards.

↓

Story Point 2 → State-wise Agricultural Production Distribution
Map showing agricultural production across Indian states.

↓

Story Point 3 → Year-wise Total Agricultural Production Bar chart showing total production trends from **1997–2021**.

↓

Story Point 4 → Trend of Agricultural Yield
Line chart showing changes in agricultural yield over the years.

↓

Story Point 5 → Comparison of Cultivated Area by Crop
Comparison of cultivated area across different crops.

↓

Story Point 6 → State-wise Cultivated Area Comparison
Comparison of cultivated area across major states.

↓

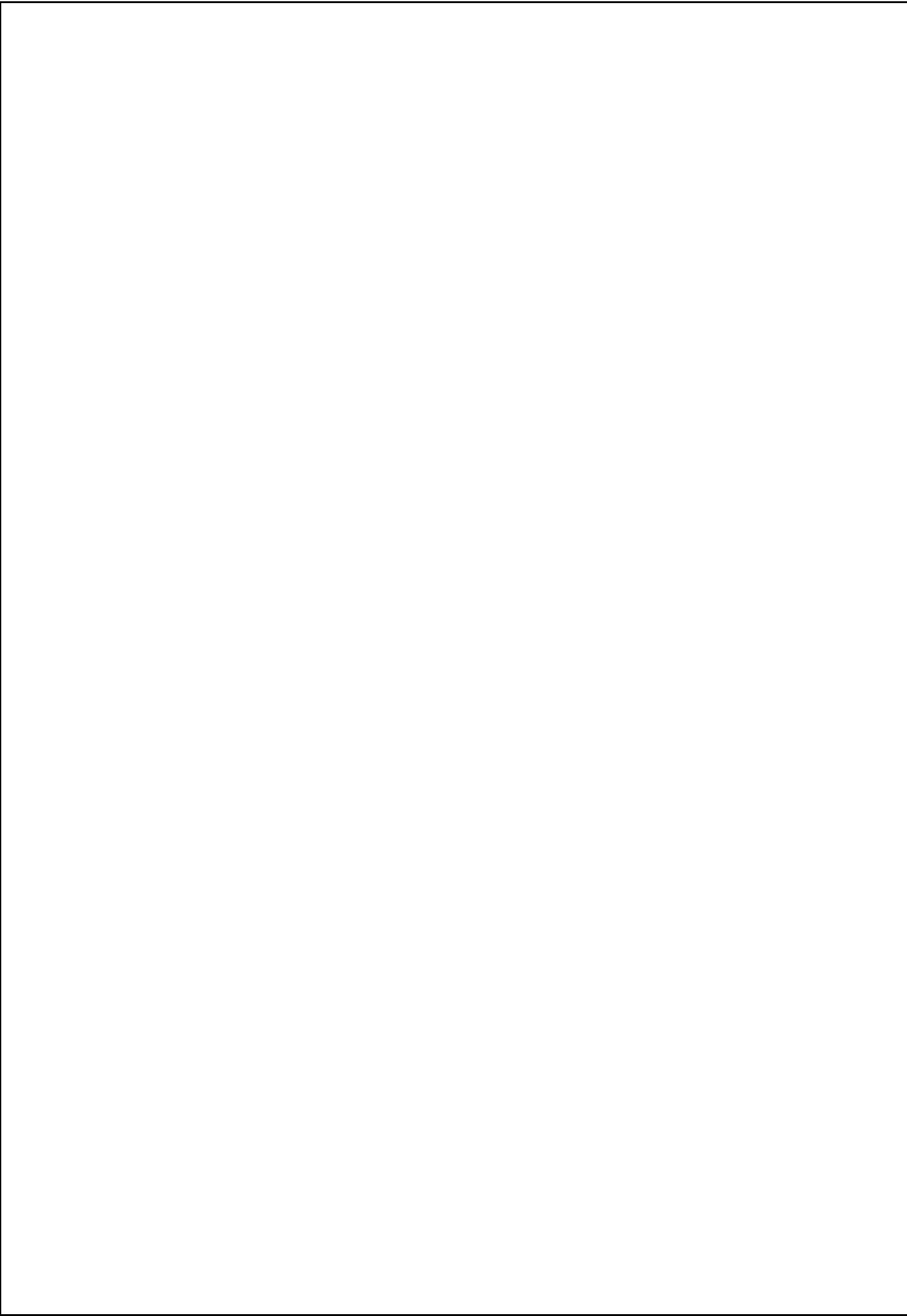
Story Point 7 → Leading State by Agricultural Production
Identifies states with higher agricultural production.

↓

Story Point 8 → Season-wise Agricultural Production Share Shows production distribution across different seasons.

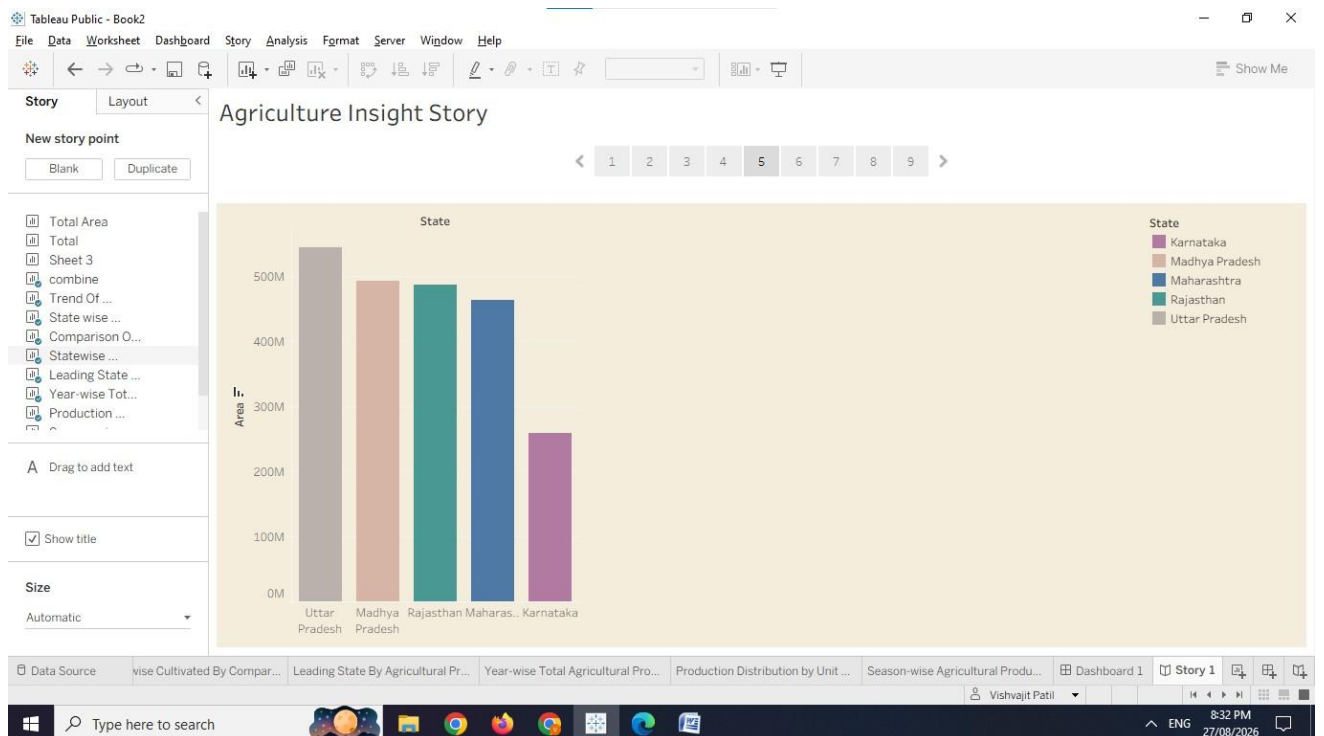
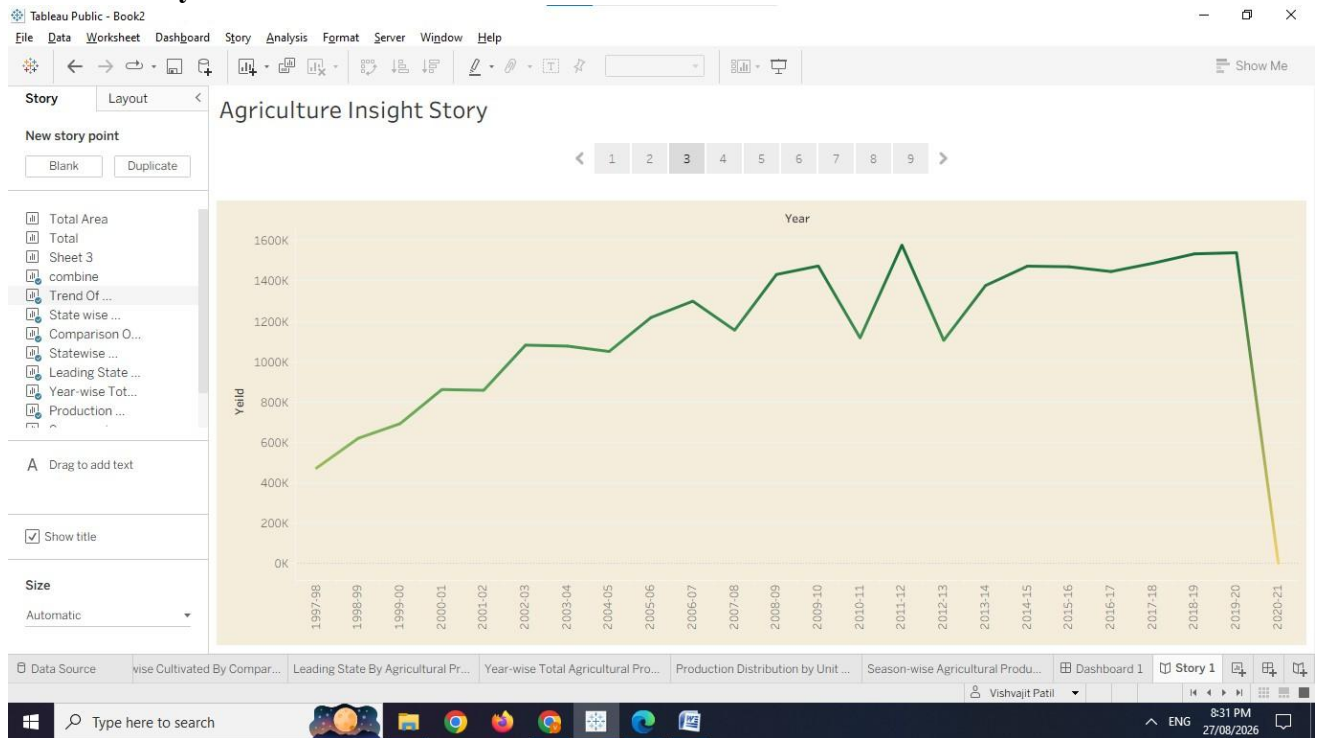
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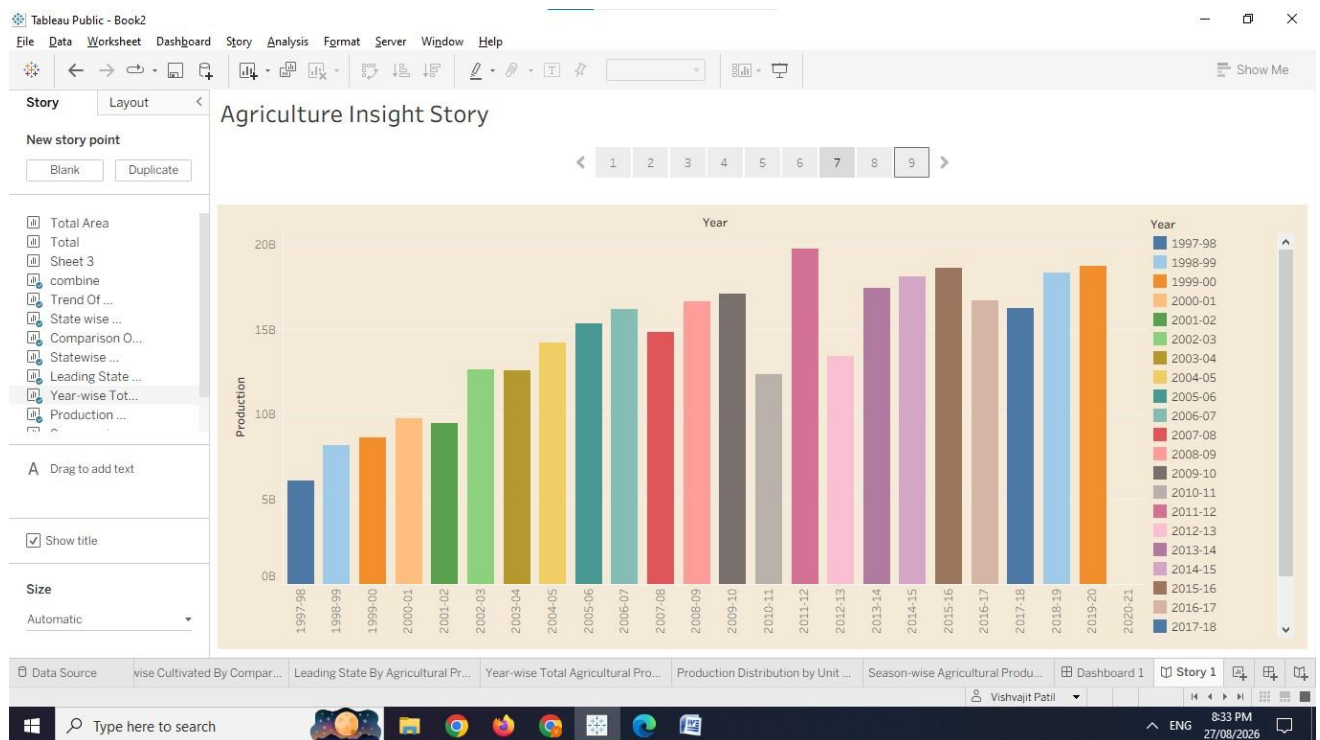
Story Point 9 → Production Distribution by Unit Type Shows production distribution by **Bales, Nuts, and Tonnes**.



Story Screenshot

Tableau Story Screenshot





Major Story Observations

- ❑ Agricultural production varies across different states.
- ❑ Year-wise analysis shows changes in total agricultural production from 1997 to 2021.
- ❑ Agricultural yield changes over the years can be analyzed through the trend visualization.
- ❑ Cultivated area varies significantly across different crops.
- ❑ States show differences in cultivated agricultural area.
- ❑ Leading states can be identified based on total agricultural production.

-
- Agricultural production varies across different seasons.
 - Production is distributed across different unit types such as **Bales, Nuts, and Tonnes**.
 - Interactive filters allow users to explore agricultural production based on **State, Year, and Crop**.
 - The Tableau Story provides an easy way to understand and compare agricultural production patterns.

Tableau Public Story Link

Link:

https://public.tableau.com/views/Book2_17877409344640/Dashboard1?:language=enUS&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

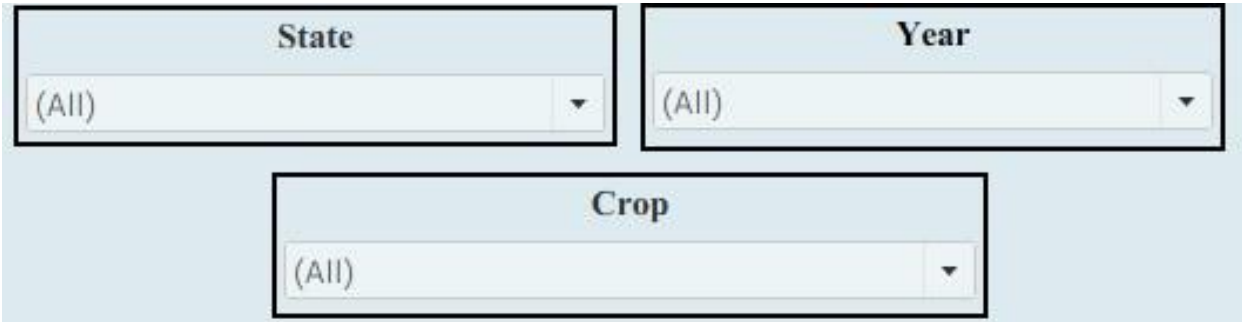
7. Performance Testing

7.1 Utilization of Data Filters

The Tableau dashboard uses interactive filters to allow users to explore agricultural production based on different dimensions.

Filter / Interactive Feature	Purpose
State	Analyze individual states
Year	Analyze production trends over time
Crop	Analyze individual crops

Screenshot Filters



7.2 Number of Calculation Fields

Number of Calculation Fields: 0

The Tableau dashboard uses the available dataset fields and standard aggregations for analysis and visualization. No separate calculated fields were created in the Tableau workbook.

7.3 Number of Visualizations

7.3 Number of Visualizations

The project contains **9 major visualizations** covering:

- State-wise Agricultural Production Distribution
- Year-wise Total Agricultural Production
- Trend of Agricultural Yield
- Comparison of Cultivated Area by Crop
- State-wise Cultivated Area Comparison
- Leading State by Agricultural Production
- Season-wise Agricultural Production Share
- Production Distribution by Unit Type
- India Agricultural Summary

Number of Visualizations:9

Sheets

-  Production
-  Yeild
-  Area
-  All Combine
-  Line Chart
-  Map Chart
-  Horizontal Chart
-  Vertical Bar ...
-  Horizontal Bar ...
-  Bar Chart
-  Dount Chart 2
-  Pie Chart

Screenshot – Tableau Worksheets

8. Conclusion / Observation

The **India's Agricultural Crop Production Analysis (1997–2021)** Tableau project provides an interactive analysis of agricultural production in India. The dashboard makes it easier to compare **states, crops, cultivated area, production, yield, seasons, and yearly trends**.

The visualizations help identify differences in agricultural production across states and crops, while year-wise analysis helps understand production and yield trends over time.

The project demonstrates how Tableau can transform raw agricultural data into meaningful visual insights and support **data-driven agricultural planning and decision-making**.

Key Outcomes

- Agricultural production can be explored interactively.

- State-wise production differences can be identified.
- Crop-wise cultivated area can be compared.
- Leading states by production can be identified.
- Seasonal production shares can be analyzed.
- Agricultural yield trends can be studied over time.
- Area and production can be analyzed together.
- Interactive Tableau dashboards provide useful agricultural insights.

9. Future Scope

- Add more recent agricultural production data.
- Include additional agricultural indicators.
- Add crop production forecasting.
- Introduce predictive analytics using machine learning.
- Include rainfall and weather data for deeper analysis.
- Add soil and irrigation-related information.
- Develop regularly updated dashboards.
- Add advanced Tableau calculations and parameters.
- Include more detailed district-level analysis.
- Publish updated versions of the dashboard on Tableau Public.

10.GitHub & Project Demo Link

GitHub Repository :

Link: <https://github.com/Vishup-hub/Nascom-final-Project.git>

Public Dashboard:

Link:

https://public.tableau.com/views/Book2_17877409344640/Dashboard1?:language=enUS&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Public Story

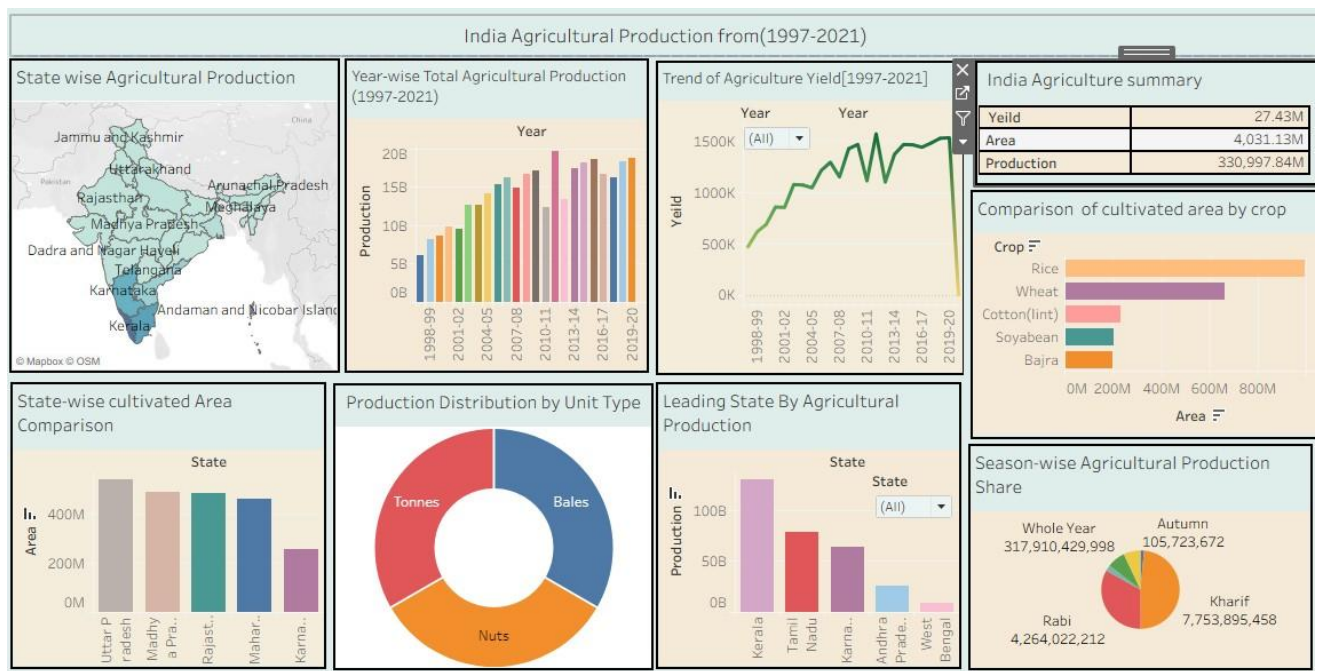
Link: https://public.tableau.com/views/Book2_17877409344640/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Project Demo Video:

Link: <https://drive.google.com/file/d/1sbSYUcGczn-ST5koLYs-mcRrRVxQIRWm/view?usp=sharing>

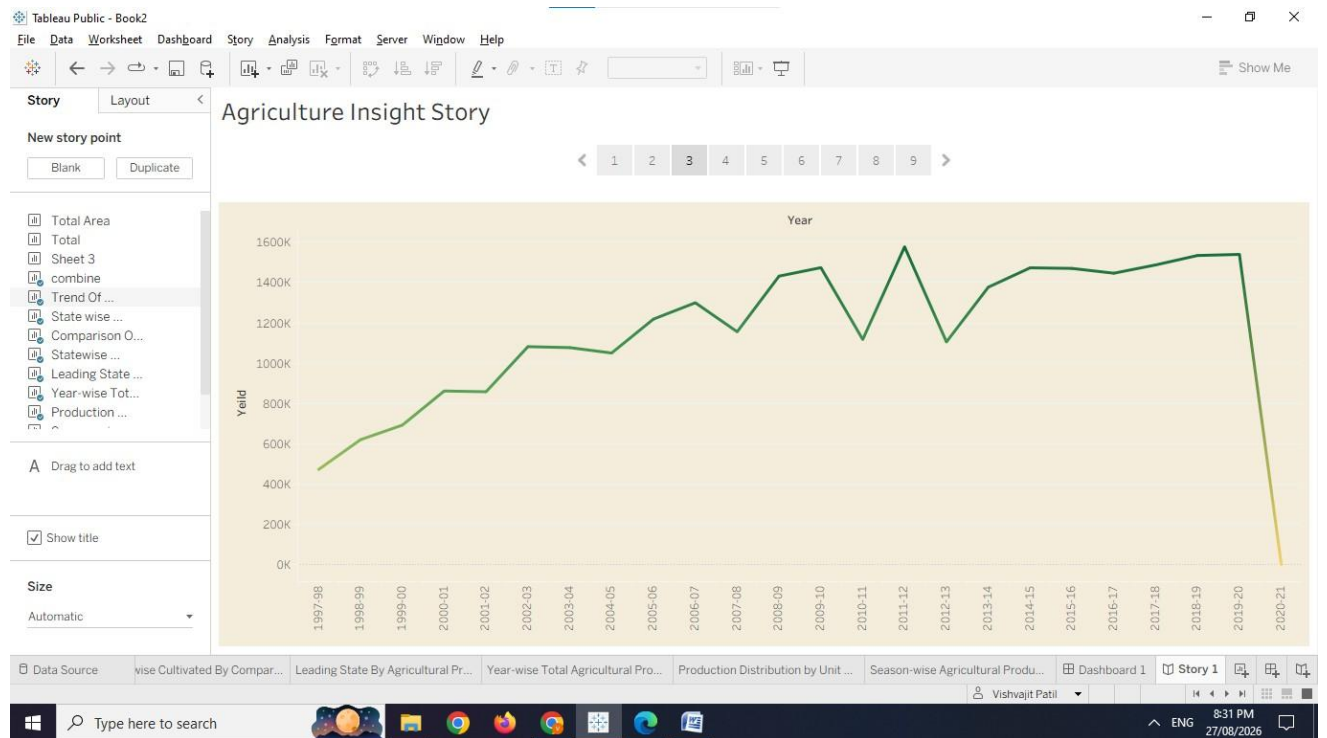
Final Dashboard Screenshot:

1.

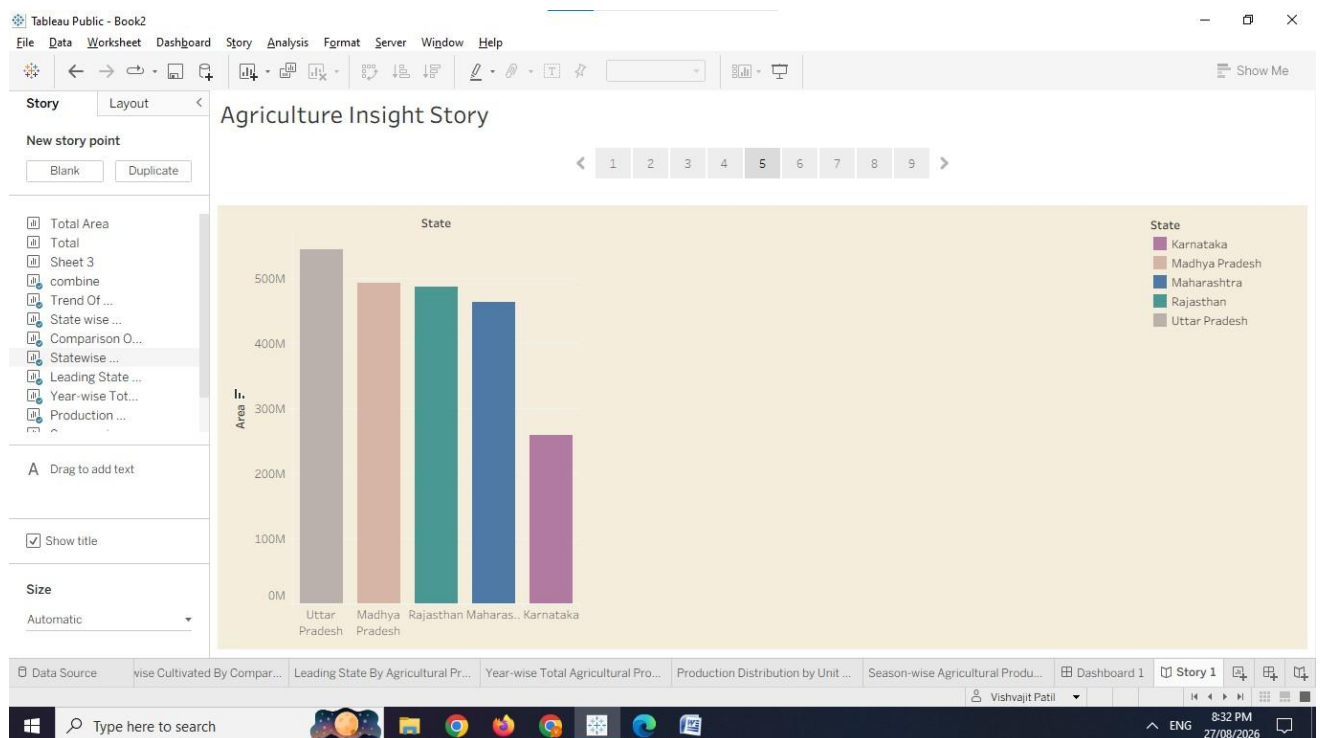


Final Story Screenshot :

1.



2.



3.

